Software Requirements Specification (SRS) Formate

Project: Online Library Management System

1. Introduction

1.1 Purpose

This document describes the requirements for the **Online Library Management System**. It is intended for developers, testers, and project stakeholders.

1.2 Scope

The system allows users to:

- Search books
- Issue and return books
- Manage users (Admin)

Benefits:

- Reduces manual work
- Faster book tracking
- Easy access to records

1.3 Definitions

Term	Meaning
Admin	System manager
User	Student/Reader
LMS	Library Management System

1.4 References

- IEEE SRS Standard
- Software Engineering textbook

1.5 Overview

This document includes system description, requirements, and test cases.

2. Overall Description

2.1 Product Perspective

This is a **standalone web-based system**.

2.2 Product Functions

- User registration
- Login/logout
- Search books
- Issue/return books
- Admin management

2.3 User Classes

User Type	Description	Skill Level
Admin	Manages system	High
User	Borrows books	Basic

2.4 Operating Environment

- OS: Windows/Linux
- Browser: Chrome, Firefox
- Internet required

2.5 Constraints

- Must use web technology (HTML, CSS, JS, DB)
- Limited project duration

2.6 Assumptions

- Users have internet access
- Database server is available

3. External Interface Requirements

3.1 User Interface

- Login page
- Dashboard
- Book search page
- Admin panel

3.2 Software Interface

- Database (MySQL)
- Backend server

3.3 Communication Interface

- HTTP/HTTPS protocol

4. System Features

4.1 User Registration

Description

Allows new users to register.

Functional Requirements

- FR1: The system shall allow users to register.
- FR2: The system shall store user details in database.

4.2 Login System

Description

Allows users to log in.

Functional Requirements

- FR3: The system shall allow login with username/password.
- FR4: The system shall validate credentials.

4.3 Book Search

Description

Users can search books.

Functional Requirements

- FR5: The system shall allow searching by title/author.
- FR6: The system shall display search results.

4.4 Book Issue

Description

Users can issue books.

Functional Requirements

- FR7: The system shall allow issuing books.
- FR8: The system shall update book availability.

5. Non-Functional Requirements

5.1 Performance

- Response time < 3 seconds

5.2 Security

- Password encryption
- Login authentication

5.3 Reliability

- System should be available 95% of time

5.4 Usability

- Simple and user-friendly interface

6. Other Requirements

- Backup system
- Logging system

7. Appendices

- Use Case Diagram
- ER Diagram

8. Test Cases

8.1 Test Case Table

Test Case ID Description Preconditions Test Steps Expected Result Actual Result Status

8.2 Test Cases**

✅ TC_01 – User Registration (Valid Data)

Field	**Details**
Description	Register new user
Preconditions	None
Test Steps	1. Open registration page 2. Enter valid details 3. Submit
Expected Result	User registered successfully
Actual Result	
Status	

✕ TC_02 – Registration with Missing Fields

Field	Details
Description	Check validation
Preconditions	None
Test Steps	Leave fields empty and submit
Expected Result	Error message shown
Actual Result	
Status	

✔ TC_03 – Login with Valid Credentials

Field	Details
Description	Successful login
Preconditions	User exists
Test Steps	Enter valid username & password
Expected Result	Redirect to dashboard
Actual Result	
Status	

✕ TC_04 – Login with Invalid Password

Field	Details
Description	Wrong password
Preconditions	User exists
Test Steps	Enter wrong password
Expected Result	Error message
Actual Result	

Field	Details
Status	

🔍 TC_05 – Search Book

Field	Details
Description	Search functionality
Preconditions	Books exist
Test Steps	Enter book name and search
Expected Result	List of matching books
Actual Result	
Status	

📚 TC_06 – Issue Book

Field	Details
Description	Issue available book
Preconditions	Book available
Test Steps	Click “Issue”
Expected Result	Book issued successfully
Actual Result	
Status	

⚠️ TC_07 – Issue Unavailable Book

Field	Details
Description	Handle unavailable book
Preconditions	Book not available

Field	Details
Test Steps	Try issuing
Expected Result	Error message
Actual Result	
Status	

8.3 Traceability Matrix

Requirement	Test Case
FR1	TC_01
FR2	TC_02
FR3	TC_03
FR4	TC_04
FR5	TC_05
FR7	TC_06
FR8	TC_07

9 Use Case Diagram/Design

9.1 Use Case Diagram

Actors

- **User (Student)**
- **Admin**

Use Cases

User:

- Register
- Login
- Search Book
- Issue Book

- Return Book

🛠 **Admin:**

- Login
- Add Book
- Remove Book
- Manage Users

9. [illegible]

External Entities:

- User
- Admin

Main Process:

- Library Management System

Data Flows:

- User → Search request
- System → Book results
- Admin → Book updates

[illegible]

[illegible]

1. User Authentication
2. Book Management
3. Issue/Return Processing

[illegible]

- User Database
- Book Database
- Transaction Records

[illegible]

Entities & Attributes

Book

- Book_ID (PK)
- Title
- Author
- Availability

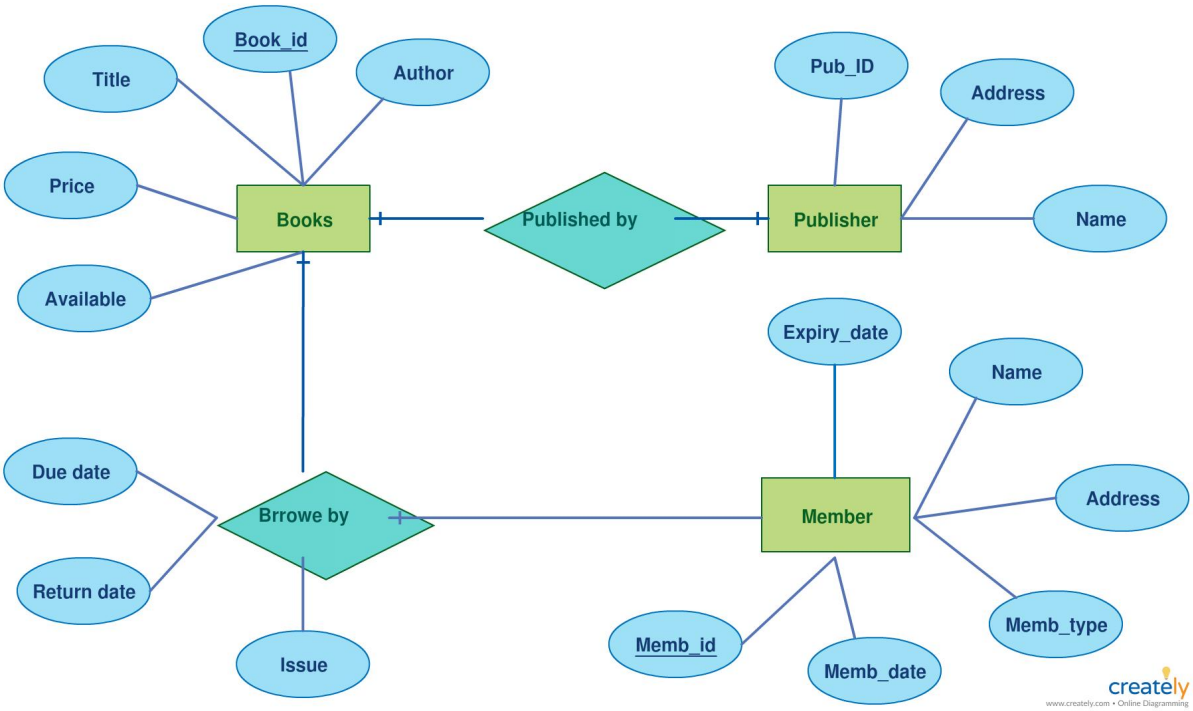
ÄUser

- User_ID (PK)
- Name
- Email
- Password

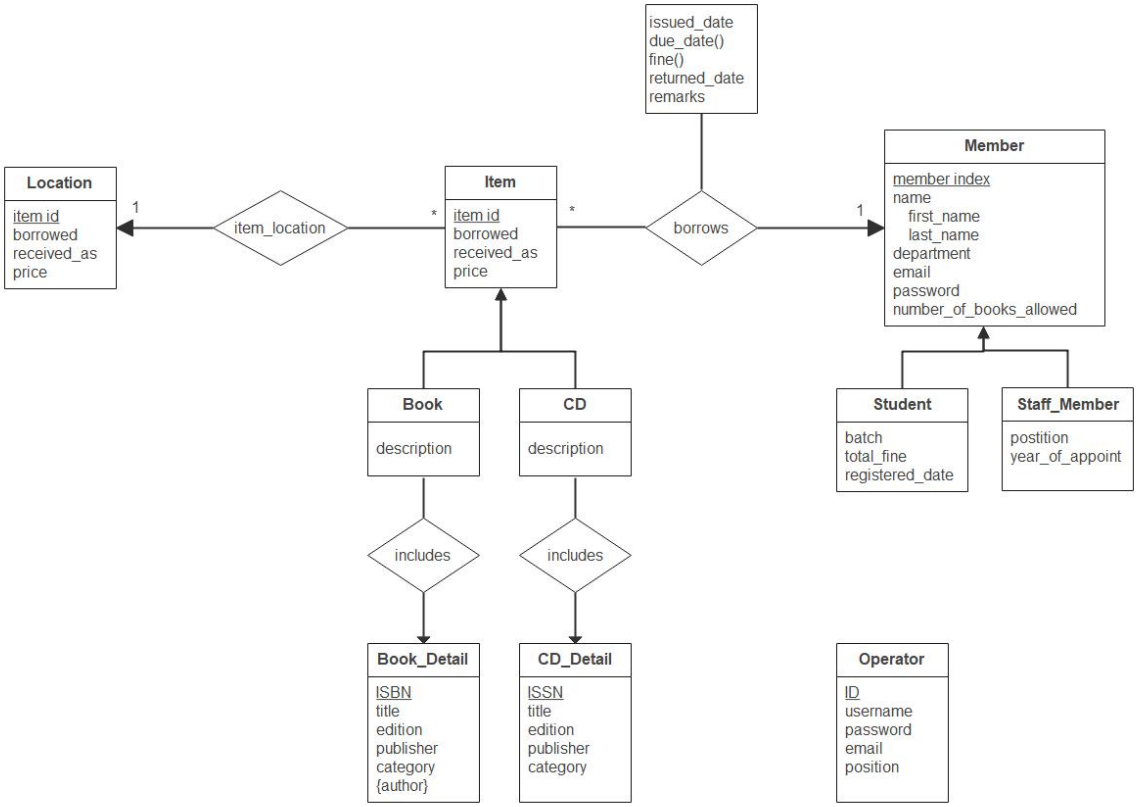
ÄTransaction

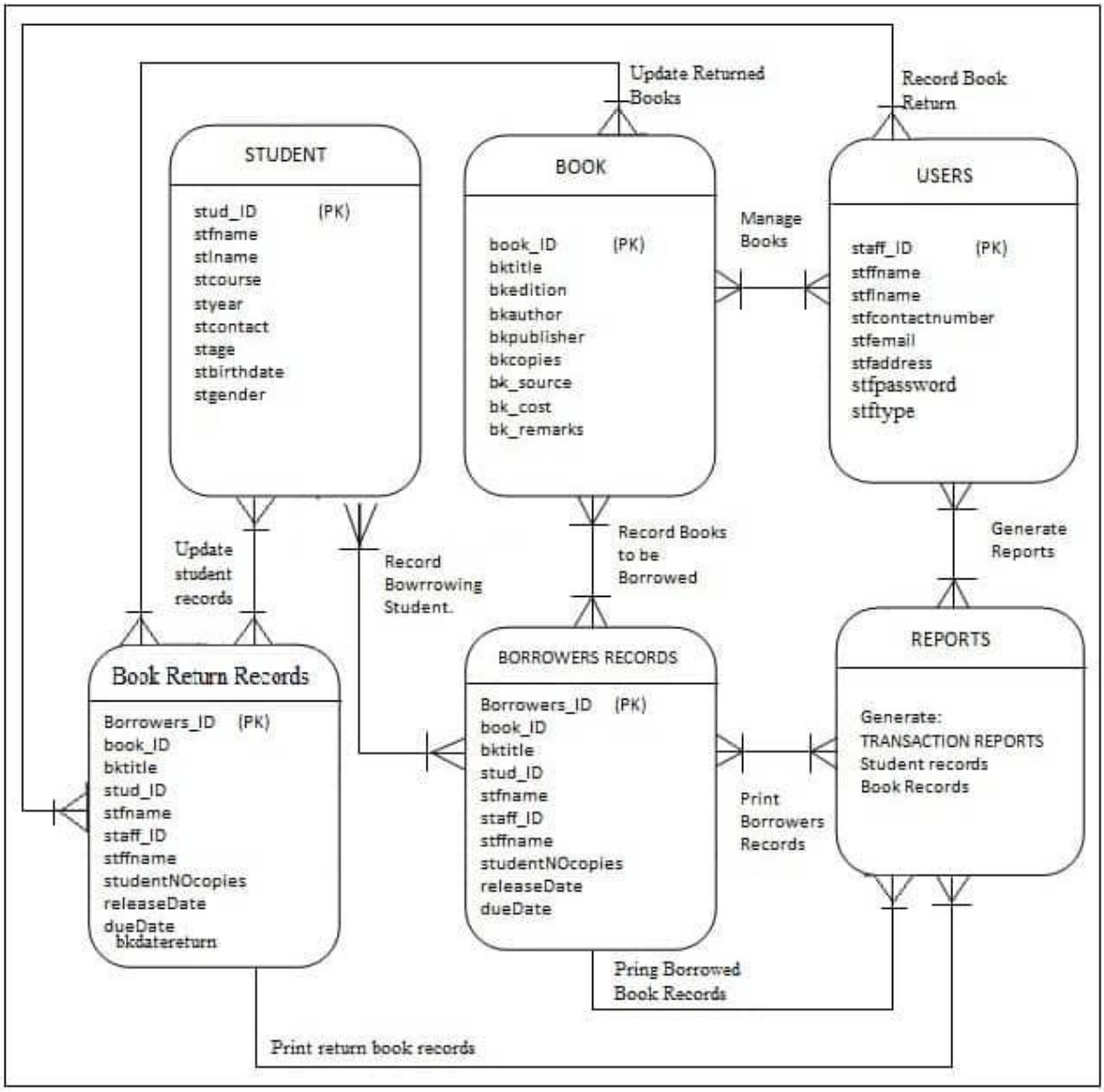
- Transaction_ID (PK)
- User_ID (FK)
- Book_ID (FK)
- Issue_Date
- Return_Date

E-R Diagram for Library Management System

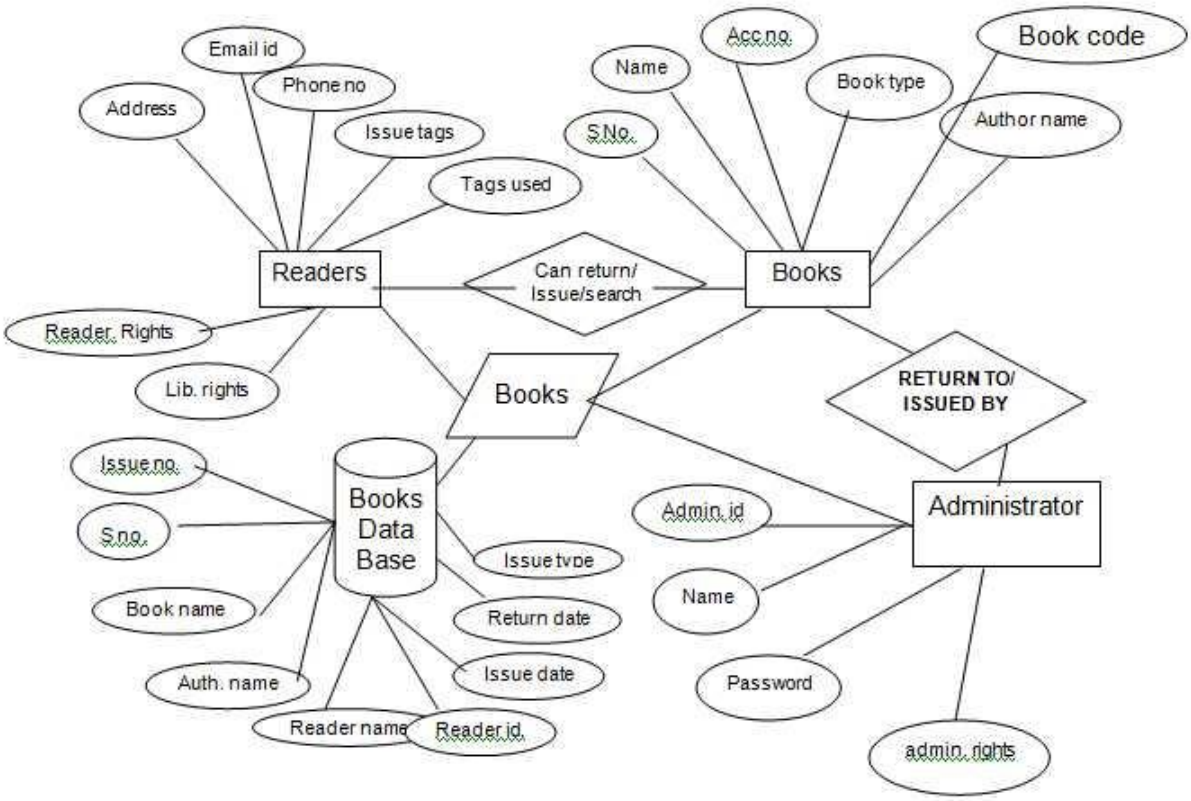


Library Management System Database Design




Update Returned Books
Record Book Return
STUDENT
stud_ID (PK)
stfname
stlname
stcourse
styear
stcontact
stage
stbirthdate
stgender
BOOK
book_ID (PK)
bktitle
bkedition
bkauthor
bkpublisher
bkcopies
bk_source
bk_cost
bk_remarks
Manage Books
USERS
staff_ID (PK)
stffname
stflname
stfcontactnumber
stfemail
stfaddress
stfpassword
stftype
Update student records
Record Bowrrowing Student.
Record Books to be Borrowed
Generate Reports
Book Return Records
Borrowers_ID (PK)
book_ID
bktitle
stud_ID
stfname
staff_ID
stffname
studentNOcopies
releaseDate
dueDate
bkdatereturn
BORROWERS RECORDS
Borrowers_ID (PK)
book_ID
bktitle
stud_ID
stfname
staff_ID
stffname
studentNOcopies
releaseDate
dueDate
Print Borrowers Records
REPORTS
Generate:
TRANSACTION REPORTS
Student records
Book Records
Pring Borrowed Book Records
Print return book records

E-R DIAGRAM



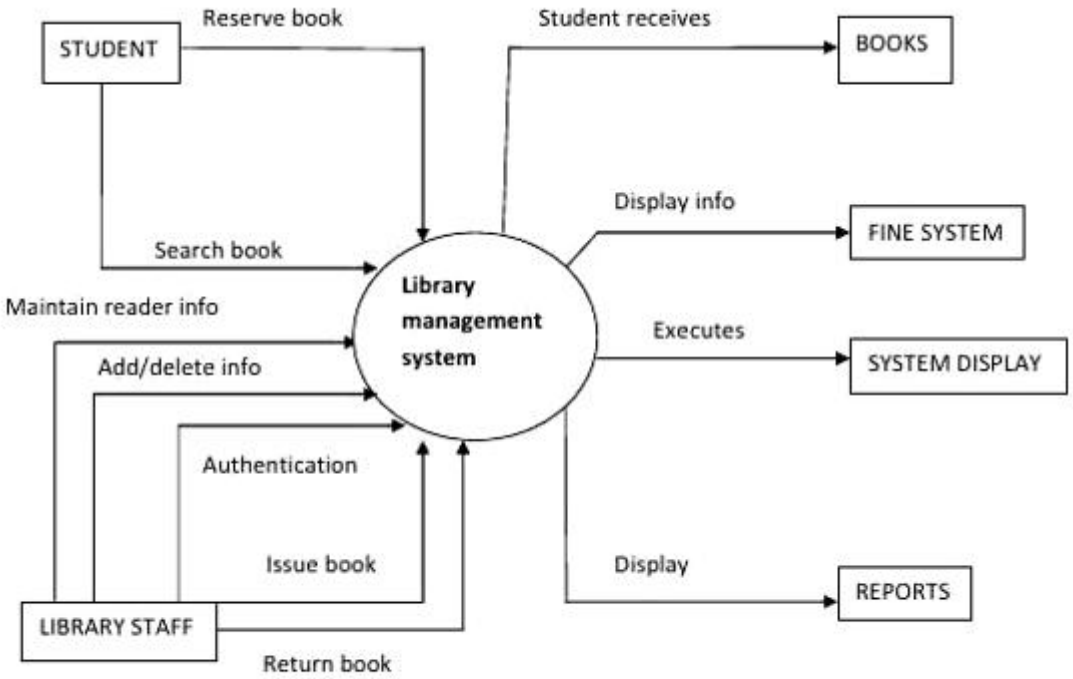
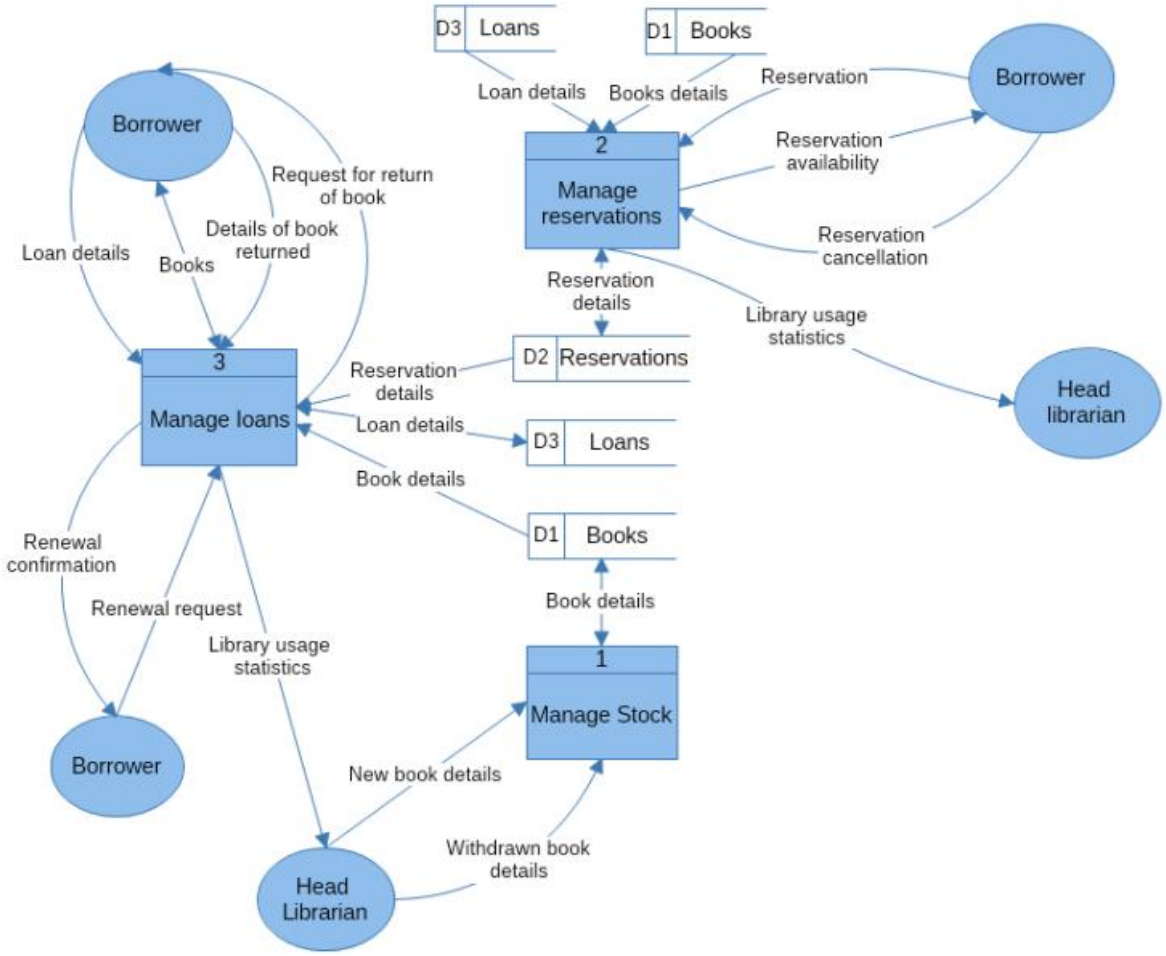
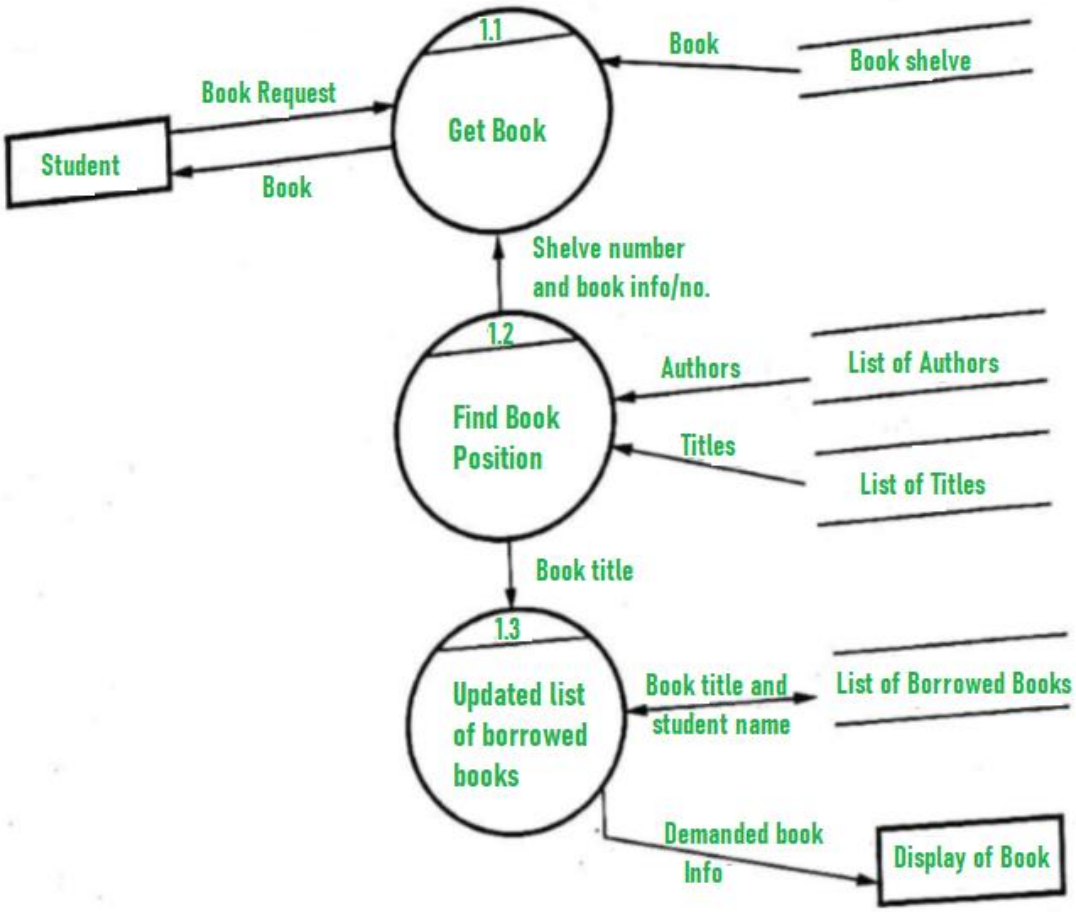


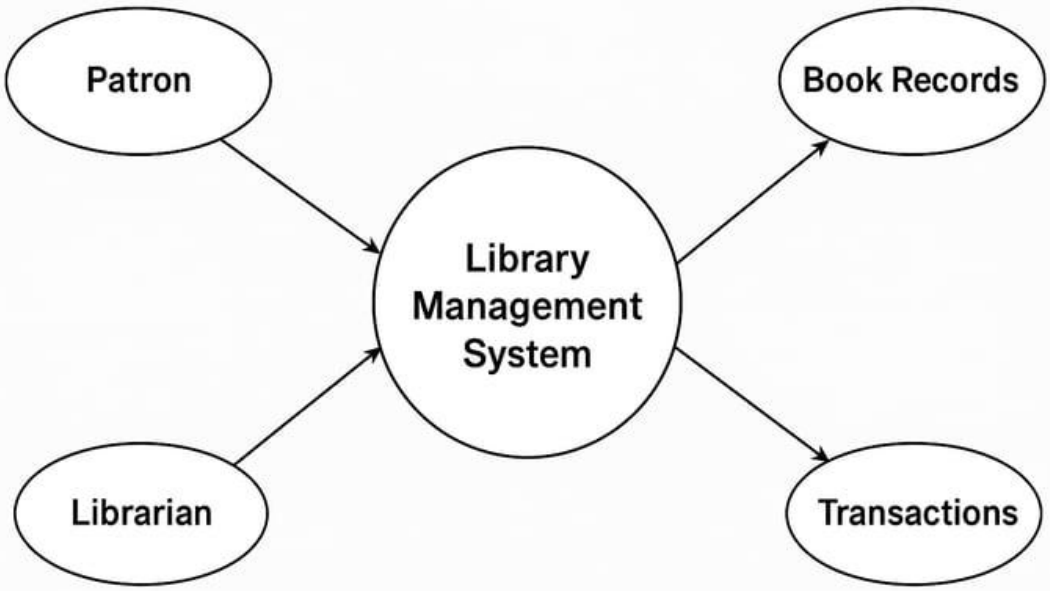
Figure 4.8 Context Level Diagram

Library System Data Flow Diagram (Level 1)

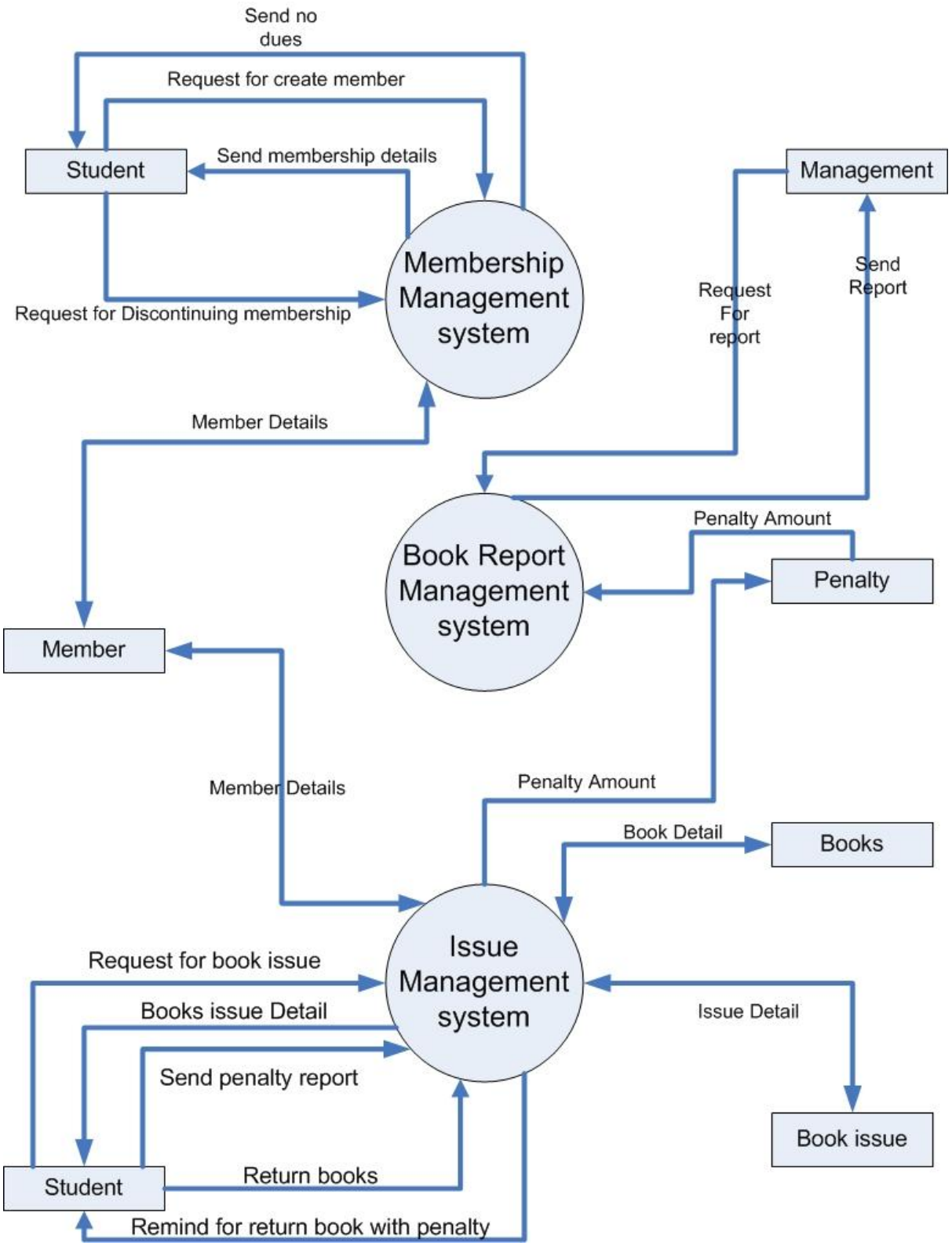


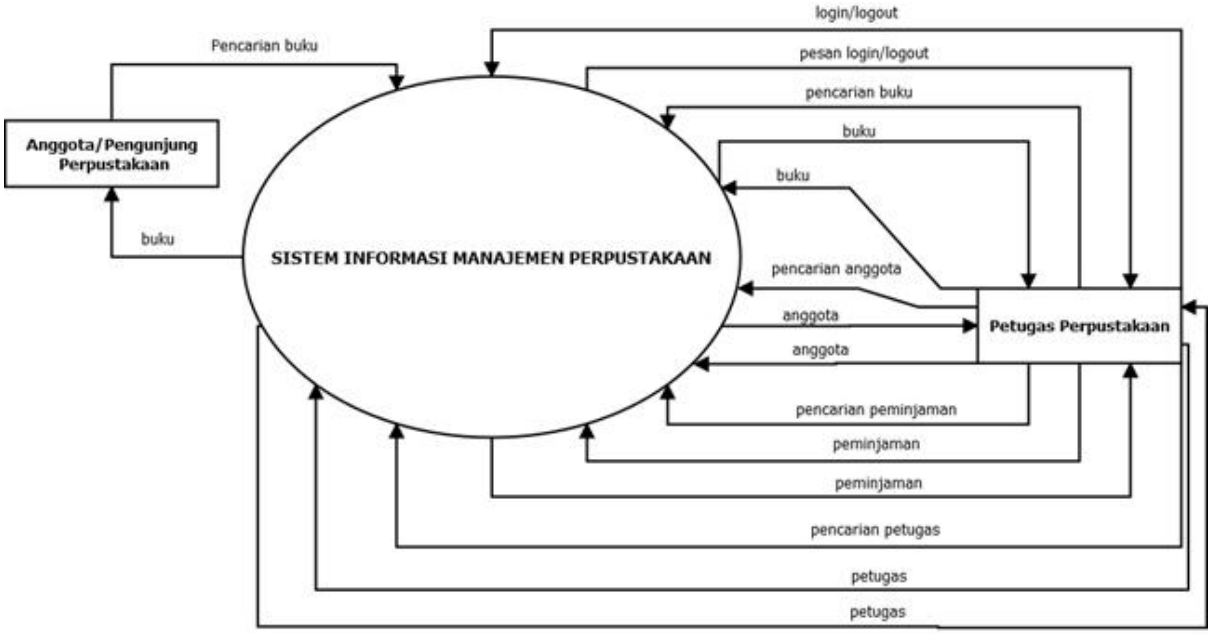


Level 2 DFD

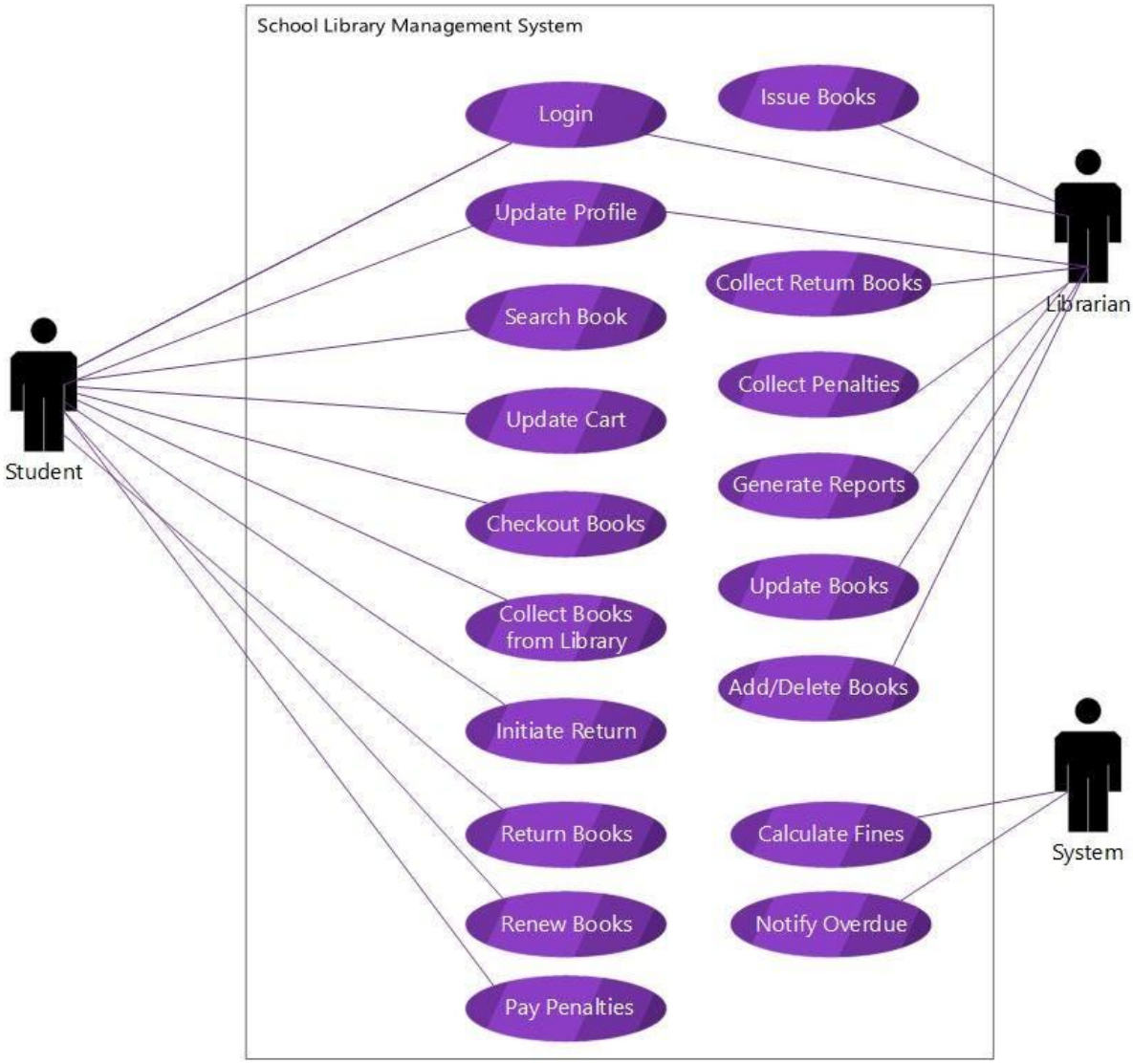


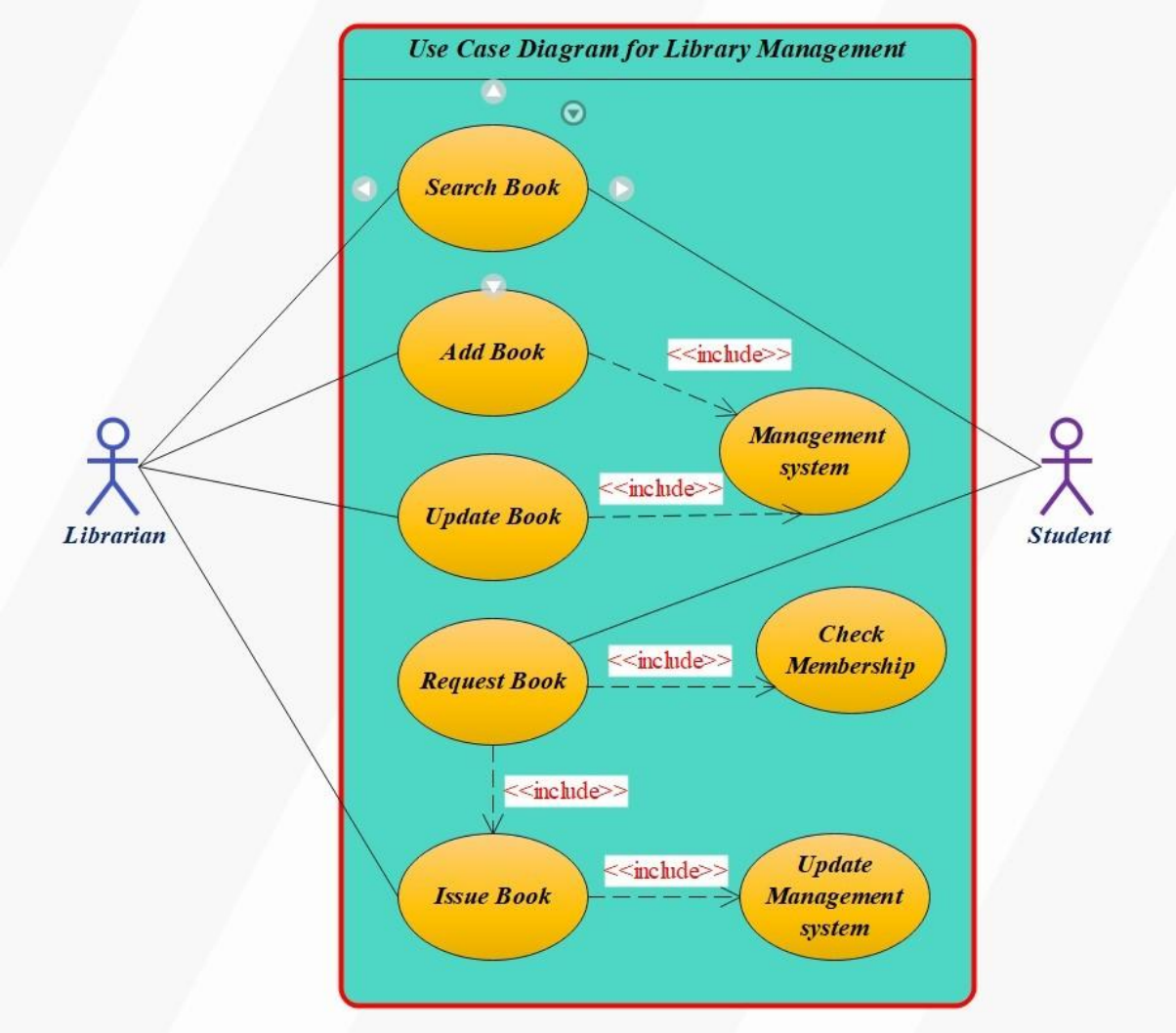
DFD Level-0


Send no dues
Request for create member
Student
Send membership details
Management
Membership Management system
Request For report
Send Report
Request for Discontinuing membership
Member Details
Penalty Amount
Book Report Management system
Penalty
Member
Member Details
Penalty Amount
Book Detail
Books
Issue Management system
Request for book issue
Books issue Detail
Issue Detail
Send penalty report
Book issue
Student
Return books
Remind for return book with penalty



Gambar 1 : DFD level 0/Diagram Konteks




Use Case Diagram for Library Management
Search Book
Add Book
Update Book
Request Book
Issue Book
Management system
Check Membership
Update Management system
<<include>>
<<include>>
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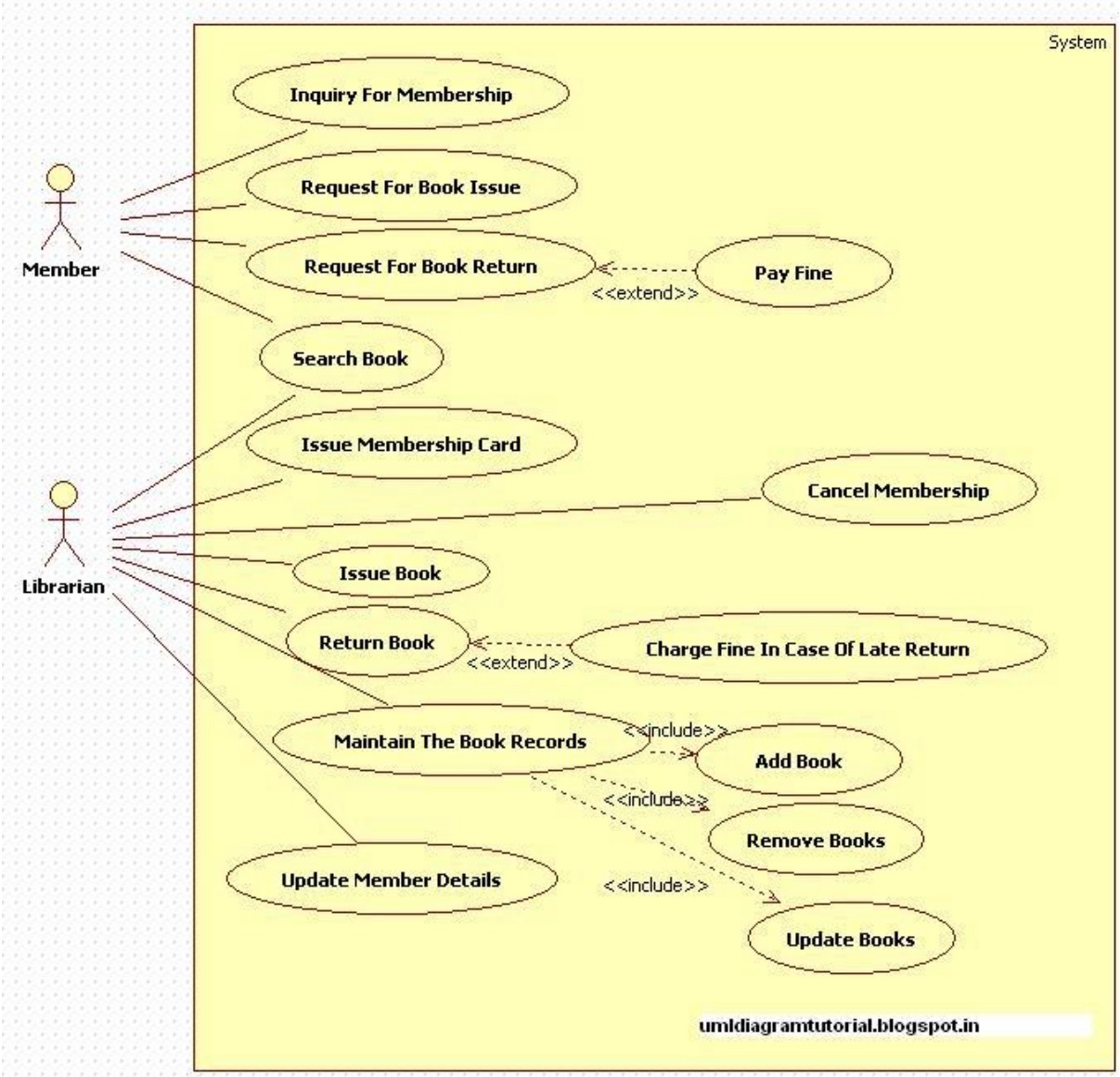
<<include>>
<<include>>
Librarian
Student


System
Inquiry For Membership
Request For Book Issue
Request For Book Return
Pay Fine
<<extend>>
Search Book
Issue Membership Card
Cancel Membership
Issue Book
Return Book
Charge Fine In Case Of Late Return
<<extend>>
Maintain The Book Records
<<include>>
Add Book
<<include>>
Remove Books
Update Member Details
<<include>>
Update Books
Member
Librarian
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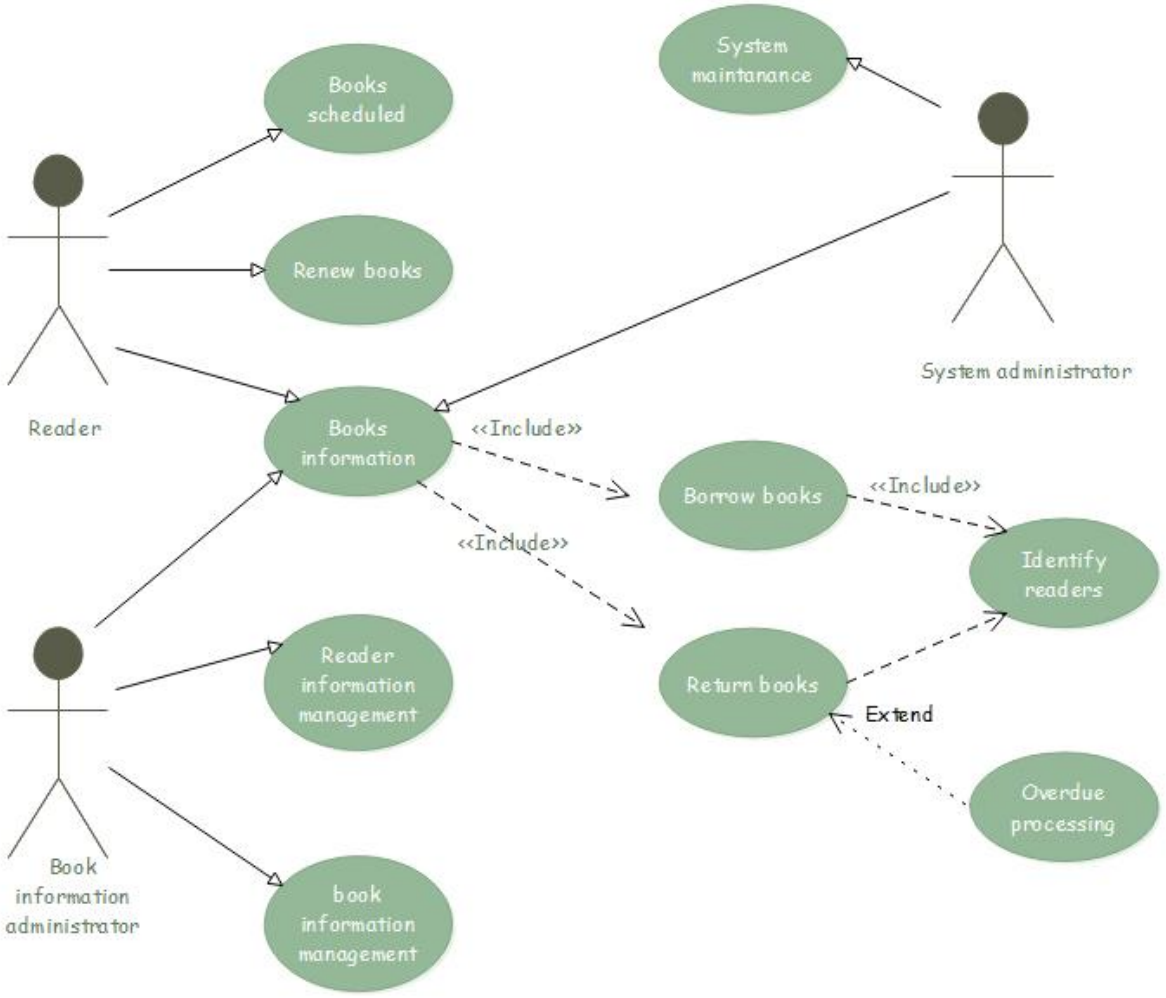
System maintanance
Books scheduled
Renew books
Reader
System administrator
Books information
<<Include>>
Borrow books
<<Include>>
Identify readers
<<Include>>
Reader information management
Return books
Extend
Overdue processing
Book information administrator
book information management